F.CSM302

1

The Sleeping Teaching Assistant

Thread, Semaphore, Mutex synchronization

Operational

: **II-4-10**

		€• ,
1	.	B242270001
2	.	B241940005
3	. - € €	B222270083
4	. € € •	B232270113
5	, . ƒ „•	B242270085

GitHub: https://github.com/B242270001/II-4-10_SleepingTeachingAssistant

… † • , 2026

f

1. „ 4

1.1 ˆ ‰ 4

1.2 • Š 4

1.3 ‹ • † €• ƒƒ €• 4

1.4 € € „ †ƒ € 4

1.5 † 5

1.6 … Š 6

1.7 Œ „ „ † 6

1.8 Event timeline: ƒƒ•, • „•, CSV 6

1.9 • €€ ƒƒ 7

1.10 ƒ €€ € 10

2. Operational „ 11

2.1 ‹ • „ • • 11

2.2 ‹ • † • 12

2.3 • • 13

2.4 •€ €•ƒƒ 13

2.5 Ž€ € € † „ • ‰ 16

…€ † ‡ •

• 1. • €• ƒ• „ € € (READY , TA) 5

• 2. S1: 20 • „ ƒ„ Š ƒƒ € 9, † 12, 3, TA ƒ„ Š † „ 7

• 3. S3: ƒ„ Š ƒƒ € 21, † 0, • Š TA 8

• 4. S2: † 78, 1. € € €€ • "ƒ„ Š ƒƒ € " • Š • (2.5- •€ „ ƒ‰) 8

• 5. PAUSED 9

• 6. STEP: WAITING FOR EVENT 9

• 7. ‹ • „ • • † •• 11

‡ • † ‡ •

ƒ € 1. • Š 4
ƒ € 2. € € „ •€ ƒƒ 5
ƒ € 3. † † • ‰ 5
ƒ € 4. … Š 6
ƒ € 5. Œ „ „ † 6
ƒ € 6. ƒ €€ € † „ € 10
ƒ € 7. Core- „ synchronization †•• 12
ƒ € 8. Core- UI • event- „ ƒƒ 12
ƒ € 9. • • 13
ƒ € 10. Š 13
ƒ € 11. € € ƒ 14
ƒ € 12. • ƒƒ „ ‰ 14
ƒ € 13. UI • † •ƒ €€ €• ƒ ƒ 15
ƒ € 14. Core €€ (UI- ƒ „) † •€ 15
ƒ € 15. UI €€ • † core- „ † • ‘ 16
ƒ € 16. UI • € € €€ • •€ 16

1. „

1.1 ˆ ‰‰ „

Sleeping Teaching Assistant (• †) ‘ ƒ„ „ • „ synchronization † ’ . „ • (TA) ’ † „• ƒ„ ƒ• . ’ €•€ TA- € €€ €, † „ •ƒ €€ €, ƒƒ € † † . ’ FIFO (€•€ € ‘ €•€) ƒ„ Š ƒƒ €.

ˆ ‘ TA † ’ †ƒ „ † thread-€€ , € „ „ Tkinter € € €€ † • ƒ € €€ • . € € € Š , ’ , €• † ƒ„ Š €€ „ • Š , • „ • ƒ„ „ .

1.2 Š † ‹

‡ • 1. Š † ‹

…	Œ
Python	3.14 (• †)
Tk	9.0 (Python- „ • € *tkinter*)
“„ „ •	macOS €€ . Windows €€ • † „
,	” ƒ„ (: threading, tkinter, queue)

Š ‹ : macOS- „ ‘ Python 3.9 €€ UI ‰ ‰ • ƒ„ Python 3.14 † Tk 9 .

1.3 €€ ‹ ‹ ‹

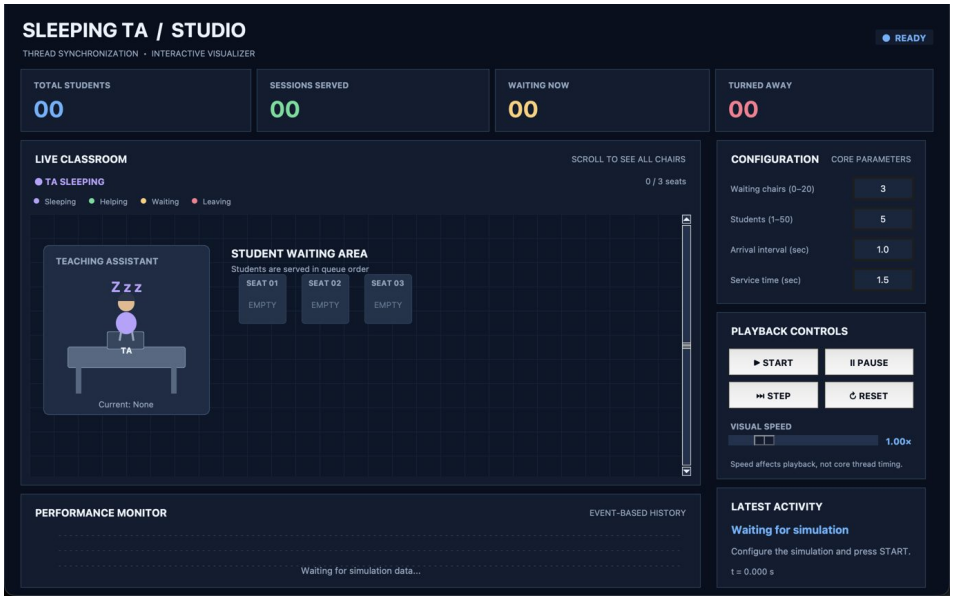
„ ƒ„. Repository- (Windows €€ python3- „ python € † Š €):

```
git clone https://github.com/B242270001/II-4-10_SleepingTeachingAssistant.git
cd II-4-10_SleepingTeachingAssistant
python3 src/ui.py
```

€ € ta_core import cd src • „ €€ python3 ui.py € .

1.4 • • •

ˆ €• €•€ • € € €€ € €. • „ ‘ **READY** † TA † „ .



1. *(READY , TA)*

‡ • 2. • ‹ ‡

‡	
€€ 4	TOTAL STUDENTS („ ’), SESSIONS SERVED (ƒ„ Š ƒƒ €), WAITING NOW (•ƒ €€ † „), TURNED AWAY (ƒƒ € †)
LIVE CLASSROOM	TA, •ƒ €€ „ , ’ . ‹ † † ƒ„ ƒƒ €€ †ƒ „ •
CONFIGURATION	‹ „ 4 †
PLAYBACK CONTROLS	START, PAUSE, STEP, RESET Š † VISUAL SPEED ƒ„ ƒƒ
PERFORMANCE MONITOR	Event †ƒ €€ €Š € €• • :ƒ„ Š ƒƒ € , •ƒ €€ † „, †
LATEST ACTIVITY	‹ƒƒ † ƒ„ „ † † „ • (t)
EVENT TIMELINE	ƒ• event- „ ƒƒ•. ” ƒƒ•, • „•, CSV † • • † „

1.5 • • €€

‹ „ €• ƒƒ €•€€ CONFIGURATION •€ € 4 . ‹ † „• ƒ• • Š • „ €• €€ RESET .

‡ • 3. • • €€ ‹Ž„

	••	Ž„	
Waiting chairs	ƒ•€	0 – 20	ƒ €€ „ . 0 † •€ Š • ƒ„, †ƒ †
Students	ƒ•€	1 – 50	— „ ’ (thread- „)
Arrival interval (sec)	• „	0.001 – 60	’ €• •
Service time (sec)	• „	0.001 – 60	TA € ’ ƒ„ Š €• •

‰ „ „ (" 3 ") € € €• • † Š € €„ (1e1 = 10) ‰ .
ƒ•€ † † • „ (3.5), • € € • †
.

1.6 • • €€

‡ • 4. • • €€

START	„ €• ƒƒ €. ƒ ‰ ƒ• ƒ € ƒƒ €
PAUSE	‹ „ ƒ ‰ . € ƒ• € € €€ • † event Š • ƒ„
STEP	ƒ ‰ ƒ• € event-€€ . Event † „• ƒ„ † **WAITING FOR EVENT** € • „ event- „ •ƒ €€ €
RESET	ƒ• , , event ƒƒ• „ € € € READY
VISUAL SPEED	ƒ € „ • (• Š 1.00˜). • • • • • , thread- „ † • • ƒ„

• • „ • († €€ †): READY, RUNNING, PAUSED, WAITING FOR EVENT, INVALID INPUT. † † INVALID INPUT Š, €• €• ƒ„.

1.7 ‘ • •

‡ • 5. ‘ • •

‘ •	••
	Ž : TA † „ (Sleeping). € € €€ "Zzz" •
	: TA ƒ„ Š † „ (Helping / ACTIVE). Ž "ƒ„ Š ƒƒ € "
	Œ : ’ •ƒ €€ † „ (Waiting). Ž "WAITING NOW"
	’ • € : ‹ ƒƒ € ’ † † „ (Leaving). Ž "TURNED AWAY"
	“ ‹ : "TOTAL STUDENTS" , READY † WAITING FOR EVENT

‹ ƒƒ € ƒ• •ƒ €€ €„, ƒ• **EMPTY** € † Š € †ƒ € •€ †€ €„ • . Performance monitor €€ • ‘ €„ (™ ƒ„ Š ƒƒ € , ™ •ƒ €€ † „, ™ †).

1.8 Event timeline: ‹, ‹ ‹, CSV

€ € „ •€ „ **EVENT TIMELINE**- core- € †ƒ• event • .

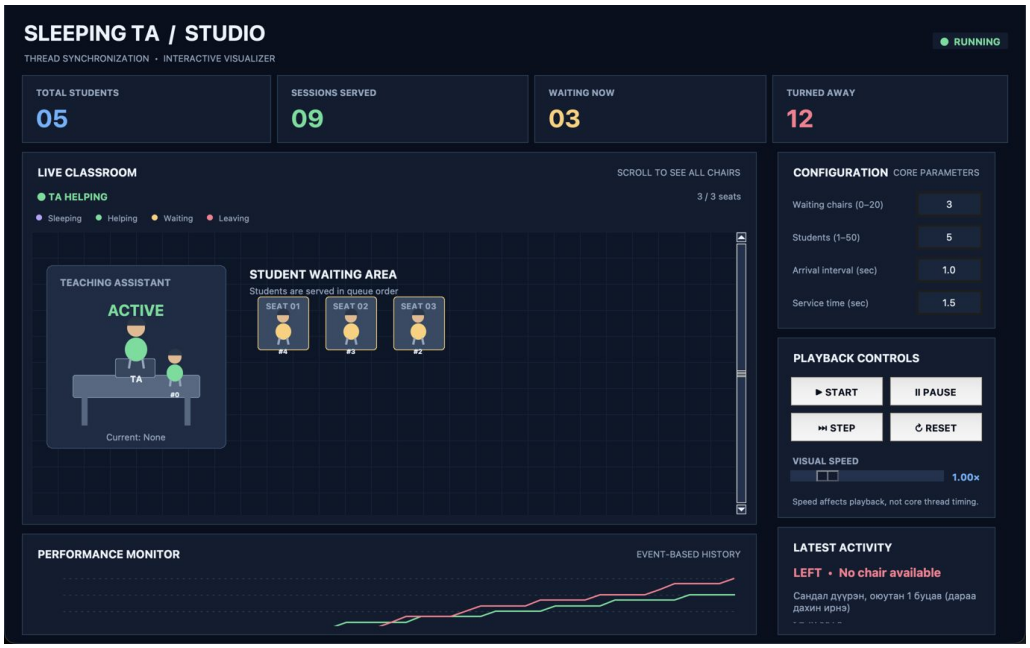
- Œ • **:** All events, TA events, Student events, Waiting, Help sessions, Turned away.
- •: • † Š •€ event- „ , ’ , „ † • „ .
- **EXPORT CSV:** event ƒƒ• „ `sleeping_ta_events_ _ .csv` € €€ • (UTF-8, Excel- Ž ƒ € ‰ •). Event † „• ƒ„ † "There are no events to export." € € € €.

1.9 Š † €€ ‹ †

• €€ ƒƒ € Š START . ‹ ƒ„ • ƒ„, „ † „ .

” 1 (S1): ‡ †

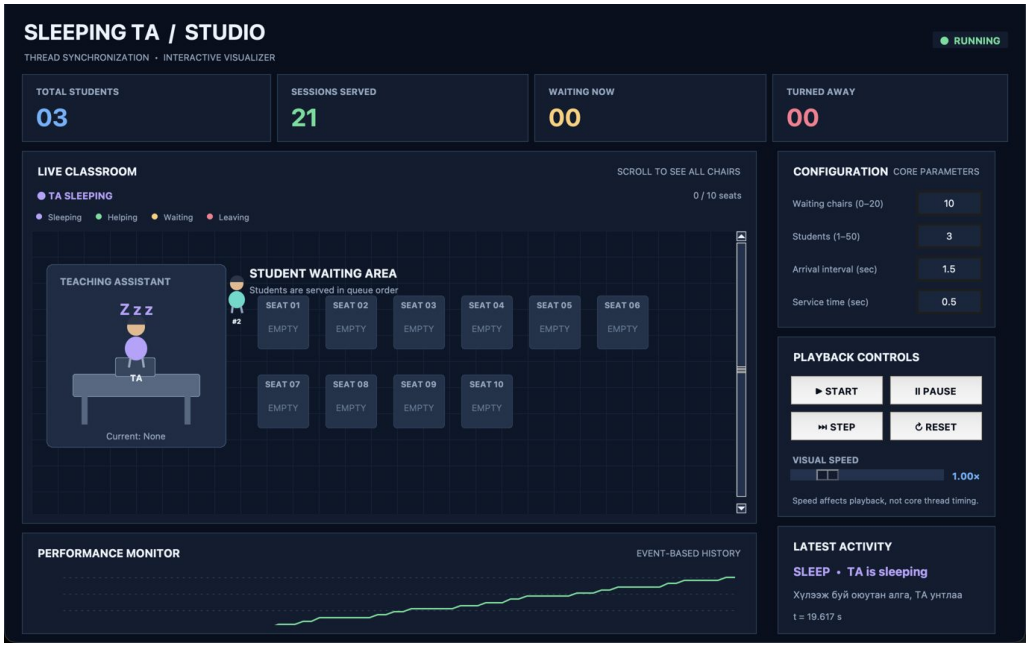
•: 3 , 5 ’ , €• • 1.0 • , ƒ„ Š €€ 1.5 • . START 20 • •ƒ €€ €. ‹ : TA €• €€ . ’ €•€ € , ƒ„ Š €. “ „ Š €€ € €€ ƒƒ Š, ‰ ’ † .



2. S1: 20 9, 12, 3, TA

” 2 (S3): ‡ ‹ ••

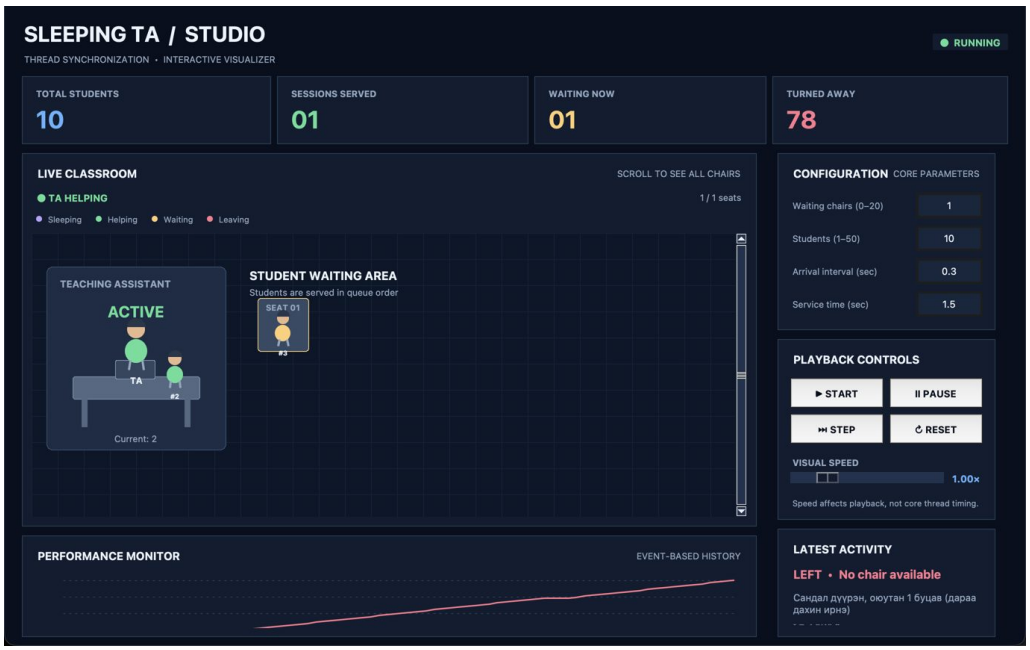
•: 10 , 3 ’ , €• • 1.5 • , ƒ„ Š €€ 0.5 • . ‹ : € Š † • ƒ„ (TURNED AWAY = 0). ’ †ƒ „ TA • .



3. S3: 21, 0, TA

” 3 (S2): ‡

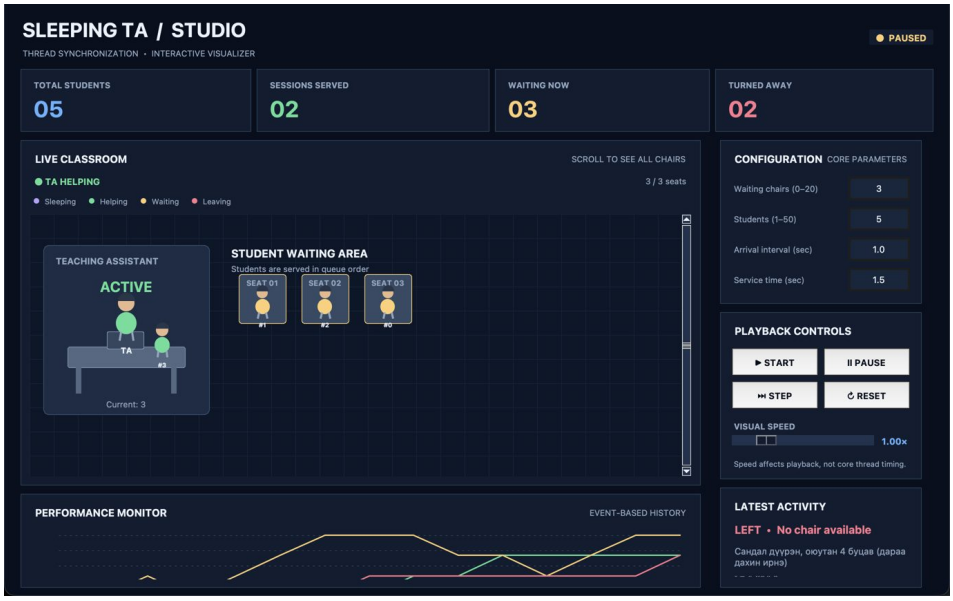
•: 1 , 10 ’ , €• • 0.3 • , ƒ„ Š €€ 1.5 • . € • ,
•€ • ’ † , 1-€€ •€ €• ƒ„.



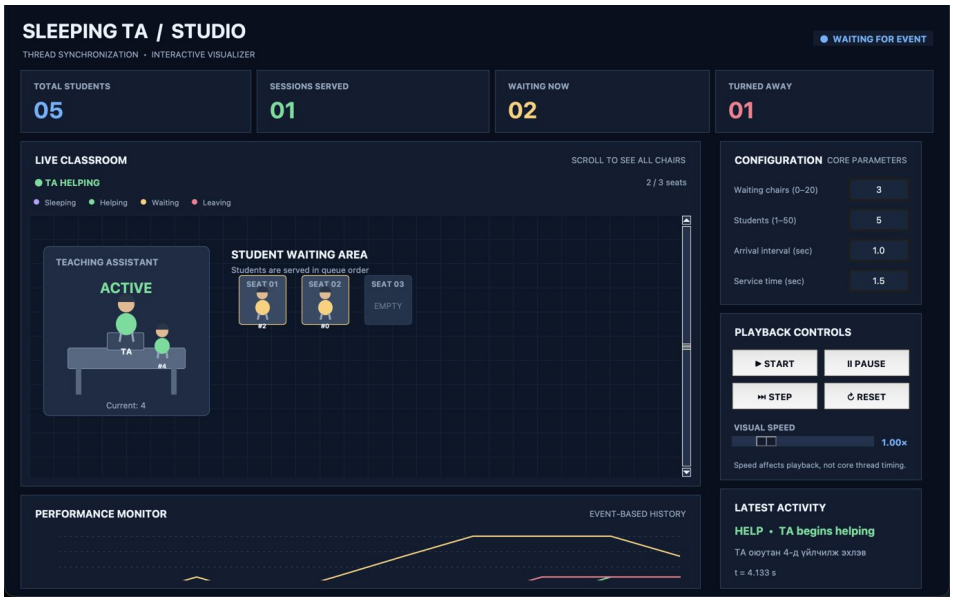
4. S2: 78, 1. " " (2.5-)

” 4 (S7, S8): PAUSE STEP

‘ S1- €„ (3 / 5 / 1.0 / 1.5). START PAUSE **PAUSED**
† , Š • † . ‘ STEP- „ • †ƒ € event . Event
† „• ƒ„ † **WAITING FOR EVENT** € • .



5. PAUSED



6. STEP: WAITING FOR EVENT

1.10 ‰

‡ • 6. ‰

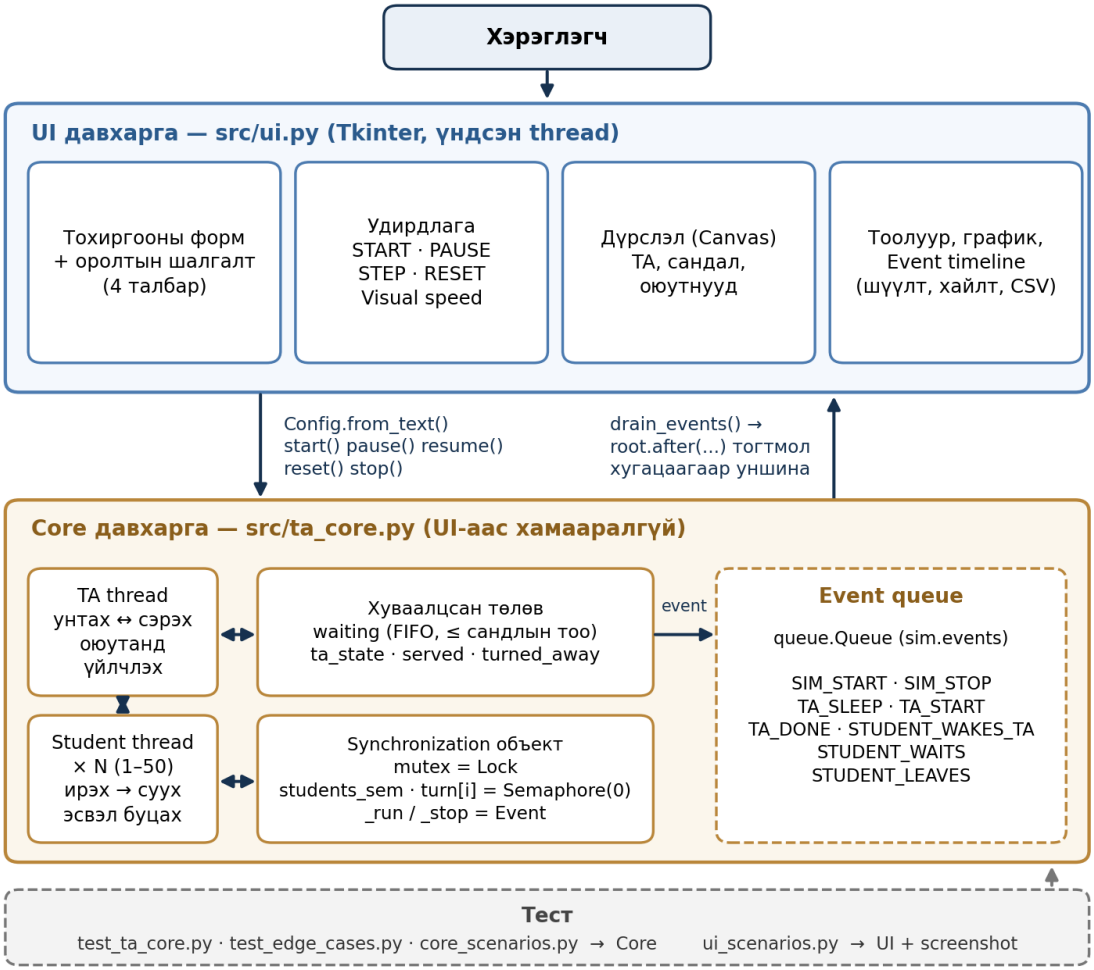
‹	/ Œ
INVALID INPUT, "‹ 0-20 • †ƒ•€ † „ ."	‹ • ‰ •€ € € (€€ ‘ 25). 0 – 20 •
INVALID INPUT, "'‹ ' ‘ †ƒ•€ † „• š „ (: 'abc')."	† • . ƒ•€ . † €€
† 0, , nan, inf	0.001 – 60 • † „• š „
• €€ €• ƒ„, tkinter	Python- Tk 9 †ƒ € €•ƒƒ € €• „ . Python 3.14
€ € † ‰ (macOS)	‹ ‘ Python 3.9- „ Tk- UI ‰ • ƒ„. Python 3.14
ModuleNotFoundError: ta_core	src • python ui.py €
core- „ † •	’ † Š „ƒ• ƒ € • Š • . 2.5- •€ „ ƒ‰
• Š † „	• €€ RESET Š , • START

2. Operational „

2.1 ‡••‰ ‹ ••,•€

‹ • ‘ • š • . **Core** (ta_core.py) ‘ TA † ’ †ƒ „ † thread-€€ , synchronization- †ƒ•€ ‘ • . Core ‘ UI- • ƒ„ ‘ † • €€ . **UI** (ui.py) ‘ Tkinter- „ ƒ € thread €€ , •€ € € Š „ ƒ„ „ core- • (`start`, `pause`, `resume`, `reset`, `stop`) .

Thread †ƒ „ ƒ„ „ event † queue.Queue- • „ €. UI ‘ `root.after(...)`-€€ € • € € €€ ‰ . „ €€ ƒ € ‘ † thread-ƒƒ „ „ • • † UI ‘ core- „ • • ƒ„.



7.

‡ • 7. Core- synchronization –•,•€€

–•,•	• •	
mutex	threading.Lock	ƒ €€ „ , TA- „ , •
students_sem	Semaphore(0)	ƒ €€ † „ ’ . TA ƒƒ €€ *wait* • „ , ’ *signal* Š € €€ €
turn[i]	Semaphore(0)	i- ’ • „ semaphore. TA ƒ„ Š ’ ƒƒ €€ •ƒ €€ €
_run	threading.Event	Set = † „ , clear = ƒ ‰ (pause)
_stop	threading.Event	ƒ• thread- „ ‰ • •

‡ • 8. Core- ‡ UI €€ ‰† ‹ event- ••

Event	**„ ‡ ‹**	**UI ‹ ‹ •**
SIM_START	‹ €•€ €	RUNNING
STUDENT_WAKES_TA	… † „ TA- ’ € €€ €	TA ACTIVE †
STUDENT_WAITS	’	Waiting € € € €, †
STUDENT_LEAVES	‹ ƒƒ € ’ †	Turned away € € € €
TA_START	TA ’ ƒ„ Š €•€ €	TA- „ • ’ , "Current: › "
TA_DONE	TA ƒ„ Š	Sessions served € € € €
TA_SLEEP	ƒ €€ † „ ’ , TA	TA SLEEPING (Zzz)
SIM_STOP	‹ ‰	€Š € € €

2.2 €€ ‹ † €€ ‹

„ †ƒ € :

```
II-4-10_SleepingTeachingAssistant/
    src/
        ta_core.py        # core: Simulation, Config, thread-
        ui.py             # Tkinter dashboard
        test_ta_core.py   #
    tests/
        test_edge_cases.py
        core_scenarios.py
        ui_scenarios.py
        screenshots/
    docs/                 # Report, Manual
    .gitignore
    README.md
```

Š ‹ ‰ 1. •:

```
git clone https://github.com/B242270001/II-4-10_SleepingTeachingAssistant.git
cd II-4-10_SleepingTeachingAssistant
```

Š ‹ ‰ 2. Python • † • (3.14 † „• š „):

```
python3 --version
```

Š ‹ ‰ 3. UI- „ •:

```
python3 src/ui.py
```

— ‡ ‰ (UI- ƒ„, ‰ • core):

```
python3 src/ta_core.py --chairs 3 --students 5 --interval 1.0 --service 1.5 --duration 10
# seed                          : --seed 42
```

Windows €€ python3- „ python (€ € py) † Š €.

2.3 •‡• † €€ ‹

ƒ• „ ƒ € • . • ‘ • š • † „ • š .

‡ • 9. •‡• † €€ ‹ , ‰ €€

— ‰	˜ € ‹	‹	€ •
`python3 -m unittest src.test_ta_core`	„ 11 •	11 • € € (*OK*)	~15 •
`python3 -m unittest discover -s tests -v`	27 edge case •	*Ran 27 tests ... OK*	~23 •
`python3 tests/core_scenarios.py`	S1–S6 • „ UI- ƒ„ †	tests/screenshots/core_results.json ƒƒ €	~2
`python3 tests/ui_scenarios.py all`	S1–S11 UI • , screenshot	tests/screenshots/ S1.png œ † S1.json œ ƒƒ €. 11 • „ € € €	~3.5

Š ‹ : `ui_scenarios.py` ‘ macOS- „ `screencapture` **„• ‹• macOS** €€ . Screen Recording ‰ † . • ƒ• • €€ € , Š .

2.4 € • ‹ ‰† ‹

€ •€ € Š , € € ƒ , • †ƒ „ •€ „ •ƒ € €€ . ‹ ƒ„ (UI- seed † „• ƒ„) †ƒ „, ƒ ƒ Š • ‘ •ƒ €€ € €„ .

‡ • 10. € •

…	••
“„ „ •	macOS 26.6.2
Python / Tk	3.14.10 / Tk 9.0
„ • †	main, commit 198e554 (release- „ € „ commit/tag- €Š €)
Core	src/ta_core.py (thread, Semaphore, Lock)
UI	src/ui.py (Tkinter dashboard), Visual speed 1.00˜
	2026.10.28

‡ • 11. € • ‡

	™	•‡• •		€ •
„ •	src/test_ta_core.py	11	11 / 11 € €	14.97 •
Edge case •	tests/test_edge_cases.py	27	27 / 27 € € (*OK*)	22.8 •
Core •	tests/core_scenarios.py	6 • (S1–S6)	JSON- •	~2
UI •	tests/ui_scenarios.py	11 • (S1–S11)	11 / 11 € €	~3.5
•		**38 • ‰ • ••‡• + 11 UI ‡••**	• •‡	

‡ • 12. Š • ‰ • ••‡• „

™	•‡•	•‡•	˜ €
test_ta_core.py	TestValidation	2	(8 •), ‰ † • ‰
test_ta_core.py	TestCorrectness	6	Invariant (, • , € € ƒ„ Š €€), Š , 0 , • † € €•, deadlock
test_ta_core.py	TestControl	3	Pause / resume / stop, reset, • start
test_edge_cases.py	TestConfigBoundaries	9	‰ , nan/inf, • ‰ „, †ƒ•€ , seed, • •
test_edge_cases.py	TestLifecycleEdgeCases	7	Stop, pause, reset- „ • † , thread
test_edge_cases.py	TestSimulationSemantics	11	FIFO, , • , TA- „ , €€ Š
•		**38**	

‡ • 13. UI ‡•• € •• • •‡• • ‹ ‹

ID		• •	•	‹
S1	“ €	€ „	3 / 5 / 1.0 / 1.5	TA €•€ . ’ €•€ € , ƒ„ Š €
S2	‹ ƒƒ €	‰	1 / 10 / 0.3 / 1.5	’ † . •€ €• ƒ„
S3	‹ • „	€ „	10 / 3 / 1.5 / 0.5	€ Š † • ƒ„, † 0
S4	0	‰	0 / 5 / 0.5 / 1.0	€ Š ƒ„ Š ƒƒ €• ƒ„, TA •€ €€
S5	–€ ’	‰	1 / 1 / 2.0 / 0.5	’ †ƒ TA € , ‘ •
S6	€€ Š	‰	20 / 50 / 0.2 / 0.5	, • ƒ„. 20- •€ €• ƒ„
S7	PAUSE / RESUME	…	3 / 5 / 1.0 / 1.5	PAUSE ƒ• • ‰ , START • ƒ € €
S8	STEP	…	3 / 5 / 1.0 / 1.5	STEP € • € event
S9	RESET	…	3 / 5 / 1.0 / 1.5	READY , †ƒ• , event 0
S10	‰ •€ € €		25 / 5 / 1.0 / 1.5	• • , €• €• ƒ„
S11	(•)		abc / 5 / 1.0 / 1.5	• • , €• €• ƒ„

‡ • 14. Core €€ (UI-) † €€ ‡ • ‹ ‰ † •

ID	•	€ • (‡•,)	‡	€•‡	€€‡	TA ‡ ‡	TA € • ‡	
S1	3 / 5 / 1.0 / 1.5	20	11	11	15	1	0	3
S2	1 / 10 / 0.3 / 1.5	20	11	434	13	1	0	1
S3	10 / 3 / 1.5 / 0.5	20	20	0	22	11	10	2
S4	0 / 5 / 0.5 / 1.0	15	0	120	0	0	0	0
S5	1 / 1 / 2.0 / 0.5	20	6	0	6	6	6	1
S6	20 / 50 / 0.2 / 0.5	20	33	2451	54	1	0	20

‡ • 15. UI ‹ ‡ • core- • • ‹ š•€€ •

ID	Core: ‡	UI: ‡	Core: €•‡	UI: €•‡	UI- ‡ / Core
S1	11	9	11	12	82%
S2	11	1	434	78	9%
S3	20	21	0	0	™
S4	0	0	120	64	™
S5	6	6	0	0	100%
S6	33	1	2451	59	3%

‡ • 16. UI ‡•• € • ‹ ‹ ‰† •

ID	‰†‡ ‹€ • /	• ‹ ‡	
S1	• €•€€ ; 20 •	• €•€€ TA . 20 • - „ ƒ„ Š ƒƒ € 9, † 12, 3, TA HELPING	€ €
S2	20 •	78, 1 (1 ƒƒ €), ƒ„ Š ƒƒ € 1 (UI •)	€ €
S3	20 •	“„ Š ƒƒ € 21, † 0	€ €
S4	15 •	“„ Š ƒƒ € 0, † 64, TA SLEEPING, 0 / 0 seats	€ €
S5	20 •	1 ’ ƒ„ Š ƒƒ „ TA • . “„ Š ƒƒ € 6, † 0	€ €
S6	20 •	RUNNING, ƒ„. 20 / 20, † 59, ƒ„ Š ƒƒ € 1 (UI •)	€ €
S7	PAUSE 4 •	PAUSED ƒ• event 15 • 15 (Š ƒ„). START- „ 30 •ƒ € ƒ € €	€ €
S8	PAUSE, STEP ˜ 5	Event 6 • 7 • 8 • 9 • 10 • 10. ‹ƒƒ „ STEP- "WAITING FOR EVENT"	€ €
S9	10 • , RESET	RESET- „ : ƒ„ Š ƒƒ € 5, † 5, event 27. RESET- „ : READY, event 0	€ €
S10	START	INVALID INPUT. "‹ 0-20 • †ƒ•€ † „ ."	€ €
S11	START	INVALID INPUT. "'‹ ' ‘ †ƒ•€ † „• š „ (: 'abc')."	€ €

Core- „ "‹ " † ‘ „ ," €€ " ‘ € Š •ƒ €€ † „ • „ ’ . UI † core- „ •€ • € €• ƒ„. € € € ƒ„: `testing_results.pdf`.

2.5 † € ‹Ž„ •

• **UI ‹ • •:** ’ † Š „ ƒ• (€€ ‘ S2, S6) UI event-ƒƒ „ • "ƒ„ Š ƒƒ € " core- „ † † • . Core- „ ‰ . S2: core 11, UI 1. S6: core 33, UI 1.

- **STEP:** event † • • • ƒ • "WAITING FOR EVENT" € • core-
  ƒ € ƒƒ € „ STEP- € €€ event € €• •
  .
- **reset(new_config):** † • „ • ‰ •€ €€ ƒ € €. UI
  • ‘ Š •€ € € Š • • ƒ „.
- **Seed:** UI- seed † † „• ƒ „, „ €€ • „ •
  † ƒ „. Seed ‘ ‰ • (--seed) † • € .
- **…• ‹ • macOS:** `tests/ui_scenarios.py` (screencapture).
- **™ •:** UI ‘ *Segoe UI* • macOS €€ •
  † .